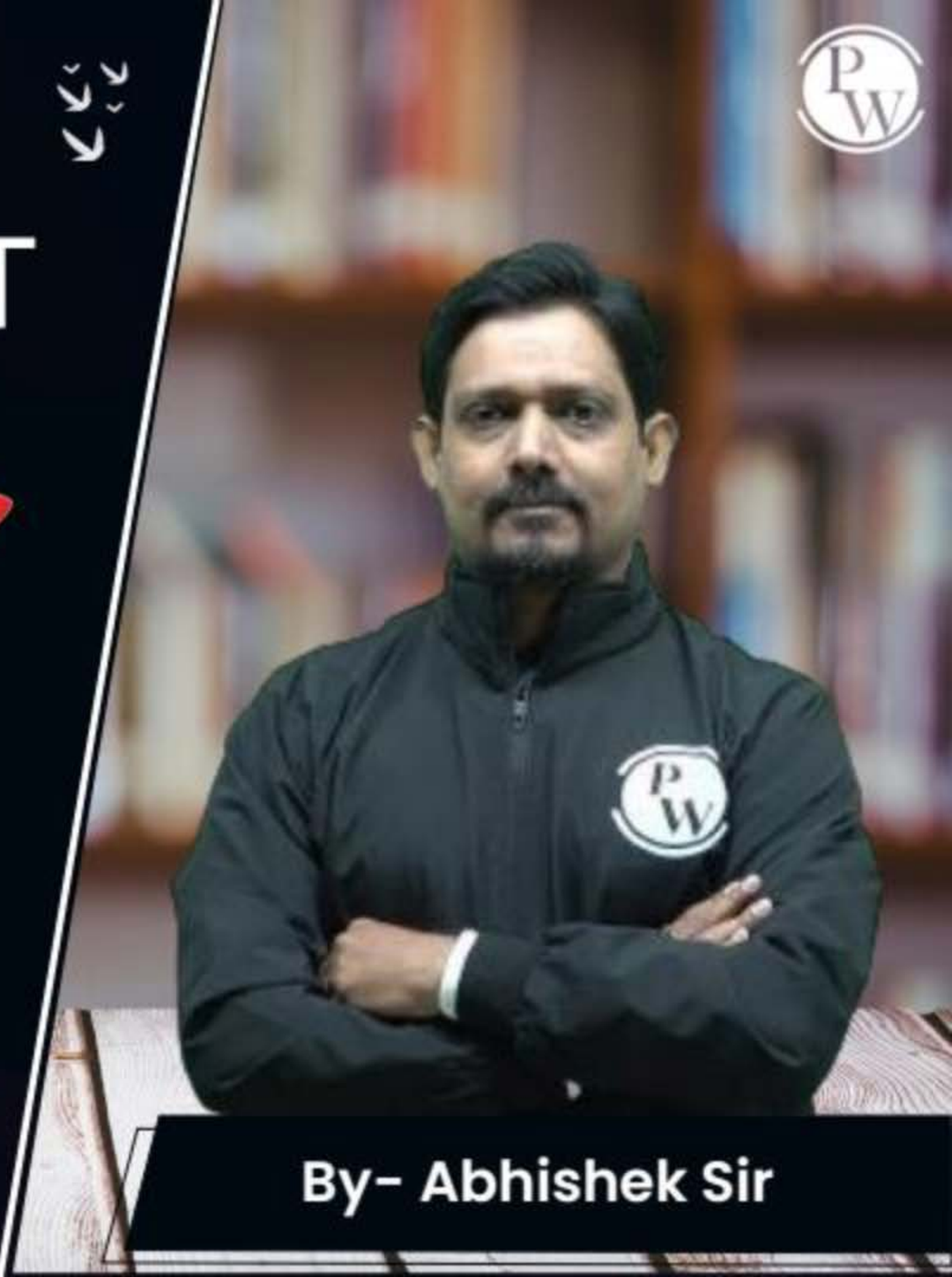
Computer Science & IT



Fundamental of C Language

Control Flow Statement

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



Topic — operators

Topic — Bitwise

Topic — Logical operators

Topic — Relational

Topic — Arithmetic

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Control flow statement

Conditional statement

if . else / Switch / Teonary

Slide



Control Flow Statement

Sequential Flow of Execution:

```
int main() {
    S1
    S2
    S3
    S4
    Si
    S6
    :
    Sn
}
```

* Each statement of program executed one after another in Sequential manner
* We want to control the Sequential flow of execution
* we want statement must be Selected based on condition

Slide



Types of Control Flow statement

1 Conditional OR Selection OR branching OR Decision

2 Iterative statement

3 Jump statement



Types of Control Flow statement

- Selection or Branching or Decision or Conditional
- Iterative statement
- Jump statements



Selection or Branching or Decision or Conditional

Selection of statement based on condition



if statement syntax

```
if (condition) {
    S1
    S2
    S3
}
```

if condition is True

the statements in the block will be executed.



if statement Example

```
int i=10, Sum=0;
if( i<=10) {
        Sum= Sum+10;
}
printf( "%d", Sum);
```

Condition - Relational

Sum will print 10

Slide



if statement Example

```
int i=9, Sum=0;
if( i<=10) {
    Sum= Sum+10;
}
printf( "%d", Sum);
```

Condition - Relational

9<=10 . false

⟶ it will print '0'

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if statement Example

```
int i=4, j=12,
if( i<5 || j>12){
    printf( At least 1 condition True");
}
printf("It will always execute");
```

→ Logical operator

$4<5 \;||\; 12>12$

$T \;||\; F = T$

At least 1 condition is true

It will always execute

Slide



if statement Example

Can we select statement if condition is false

Slide



If else Statement

```
int main(){
    s1;
    if(condition/expression){
        s2;
        s3;
    }
    else {
        s4;
        s5;

    }
    s6;
}
```

branching

true → S_2, S_3

if condition is false

then branch to else



If else Statement

```
#include<stdio.h>
int main(){
    int i=4, j=10;
    if(i>1 && j<20)
        printf("Condition satisfied");
    else
        printf("Condition Not satisfied");
return 0;
}
```

if blocking is Not done then only 1 statement is part of selection

output

Condition satisfied

Slide



If else Statement

```
#include<stdio.h>
int main(){
    int i=4, j=21,
    if (i>1 && j<20)
        printf("Condition satisfied");
    else
        printf ("Condition Not satisfied");
return 0;
}
```

if blocking is Not done then only 1 statement is part of selection

output

Condition Not satisfied

Slide



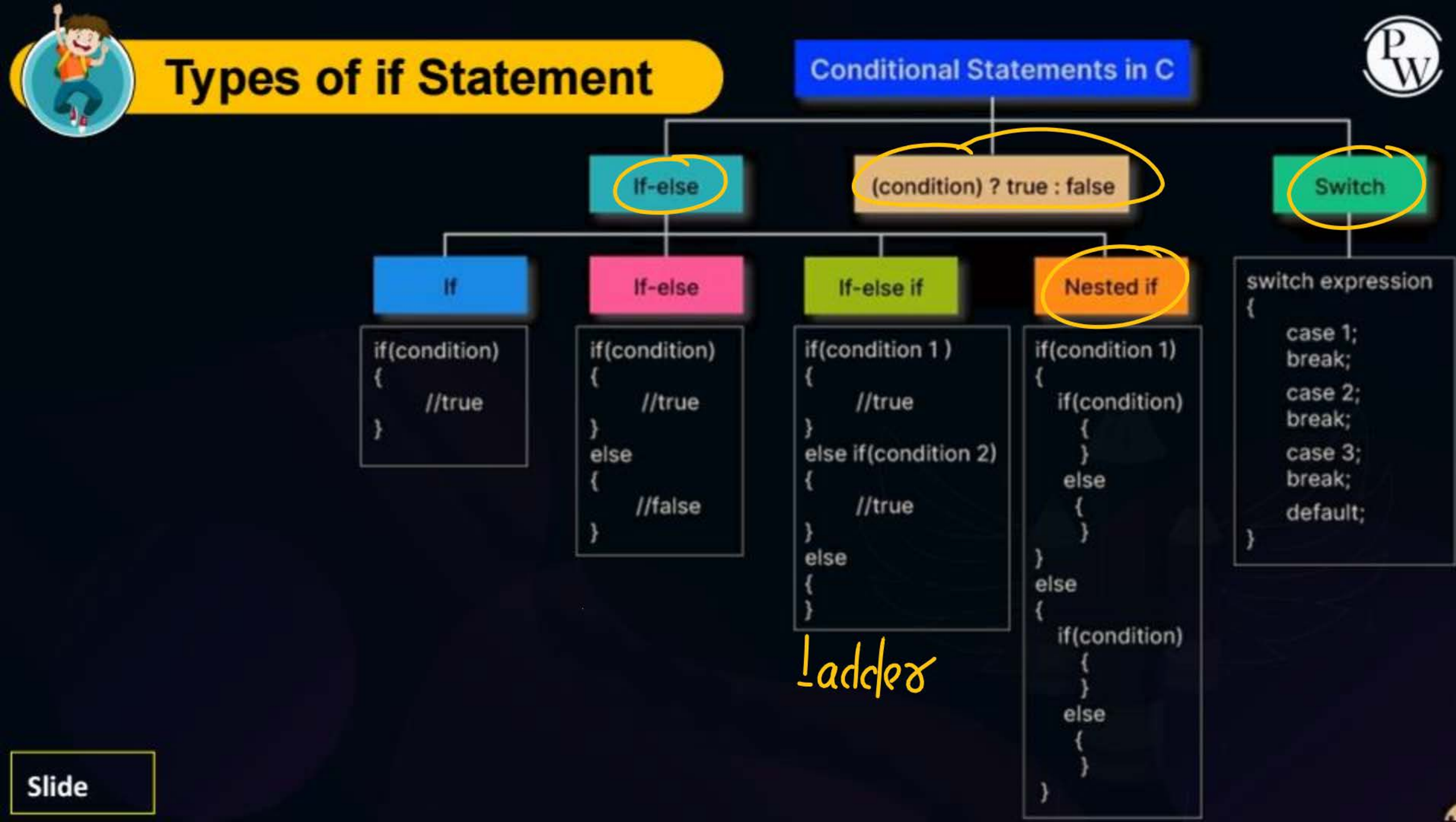
If else Statement

C Language true — Non Zero –

false — Zero

Negative is also True

Types of if Statement



Slide



Nested if else Statement

if within if block if another if-else block is present

the we call it as Nested statement



Nested if else Statement

```
#include<stdio.h>
int main() {
    int a=20
    if (a<25) {
        if(a>20)
            printf("a is Less than 25 and greater than 20");
        else
            printf("a is less than 25 but Not greater than 20");
    }
    return 0;
}
```

Nested if statement

Slide



If else ladder

```
#include<stdio.h>
int main(){
    int a=15;
    if (a<10){
        printf("a is less than 10");
    }
    else if(a<20)
        printf("a is less than 20");
}
```

output a is less than 20

Slide



Ternary operator ? :

3 operands are present

$$expr_1 \ ? \ expr_2 \ : \ expr_3$$

$expr_1$ is conditional statement value is true or false

if true then $expr_2$ evaluated

if false then $expr_3$ evaluated

Slide



Ternary operator

```
# include<stdio.h>
int main(){
    int a;
    a = 4>5 ? 20 : 30;
    printf("%d", a);      this will print 30
    a = 10<20 ? 40 : 30; // a = 40
    printf("%d", a);     // It will print 40
    return 0;
}
```

a = 30 because 4>5 is false

Associativity
2
precedence

Slide



2 mins Summary

Topic — Control flow types

Topic — if, if-else

Topic — if else if, Nested if

Topic — Ternary operator

Topic

Slide

THANK - YOU

